

TED(15)4092

Reg No.....

(REVISION-2015)

Signature.....

FOURTH SEMESTER DIPLOMA MODEL EXAMINATION IN POLYMER
TECHNOLOGY

LATEX TECHNOLOGY

(Maximum marks - 100)

(Time - 3Hrs)

PART - A

(Maximum marks - 10)

I Answer the following questions in one or two sentences. Each carries two marks.

1. State the functions of wetting agents in latex compounding.
2. List out any four dipped goods.
3. State the classification of casting.
4. Define gelation of Latex.
5. State the principle of latex extrusion. (5x2=10)

PART - B

(Maximum marks - 30)

II Answer any five of the following questions. Each carries six marks

1. State any four principles of latex compounding
2. Define synthetic latex and state its four applications.
3. Describe heat sensitised dipping process.
4. Describe rotational casting.
5. Distinguish between Dunlop and Talalay process.
6. Describe spraying method of method of carpet backing
7. Describe any one method for latex spreading. (5X6=30)

PART - C

(Maximum marks - 60)

(Answer one full question from each Module. Each full question carries 15 marks)

MODULE - I

- III (a) Compare latex compounding with dry rubber compounding. (6)
- (b) Describe the method of preparation of ZnO dispersion using Ball mill (6)
- (c) Illustrate the role of activators in latex compounding (3)

OR

- IV (a) Explain the principles of latex compounding (6)
- (b) Explain the preparation of Oil emulsion (6)
- (c) State any three characteristics of synthetic latices. (3)

MODULE - II

- V (a) Explain straight dipping process. State the relation between thickness of the deposit formed and number of dips. (7)
- (b) Explain the preparation of plaster of Paris mould for casting ball. (8)

OR

- VI (a) Explain the process of manufacture of finger cap with formulation. (7)
- (b) Explain the common defects, their remedies of dipped goods. (8)

MODULE III

- VII (a) Explain Dunlop batch process (6)
- (b) Explain "paper coating method" for treatment of paper (6)
- (b) State any four applications of latex foam products (3)

OR

- VIII (a) Explain the production of Coir foam. (6)
- (b) Explain the method of application of latex in cement compositions. (6)
- (c) Compare Dunlop and Talalay process. (3)

MODULE - IV

- IX (a) Explain the manufacturing process of latex thread with formulation. (8)
- (b) Compare latex based adhesives with solvent based adhesives (7)

OR

- X (a) Explain any two manufacturing process of latex tube. (8)
- (b) Compare latex spreading and impregnation (7)